

Multiple Choice

1. **Answer:** A

Options: A) 0.0125 equiv/liter; B) 0.060 equiv/liter; C) 0.050 equiv/liter; D) 0.100 equiv/liter; E) 0.125 equiv/liter

Note: Correct option: A - 0.0125 equiv/liter

2. **Answer:** E

Options: A) 7.39×10^{-5} ; B) 9.12×10^{-5} ; C) 3.43×10^{-5} ; D) 6.28×10^{-5} ; E) 1.71×10^{-5}

Note: Correct option : E - 1.71×10^{-5}

2. **Answer:** E

Options: A) $8.110^6 \text{ m} \cdot \text{s}^{-1}$; B) $9.410^7 \text{ m} \cdot \text{s}^{-1}$; C) $6.210^7 \text{ m} \cdot \text{s}^{-1}$; D) $1.110^8 \text{ m} \cdot \text{s}^{-1}$; E) $3.710^7 \text{ m} \cdot \text{s}^{-1}$

Note: Correct option: E - $3.710^7 \text{ m} \cdot \text{s}^{-1}$

3. **Answer:** A

Options: A) Bond order of He₂ is 0 and He₂⁺ is 0.5; B) Bond order of He₂ is 0.5 and He₂⁺ is 0; C) Bond order of He₂ is 0.5 and He₂⁺ is 0.5; D) Bond order of He₂ is 0 and He₂⁺ is 0.5

Note: Correct option : A - Bond order of He₂ is 0 and He₂⁺ is 0.5

3. **Answer:** A

Options: A) $[\text{H}_2\text{SO}_3] > [\text{HSO}_3^-] = [\text{SO}_3^{2-}]$; B) $[\text{HSO}_3^-] > [\text{SO}_3^{2-}] > [\text{H}_2\text{SO}_3]$; C) $[\text{SO}_3^{2-}] > [\text{H}_2\text{SO}_3] > [\text{HSO}_3^-]$; D) $[\text{HSO}_3^-] = [\text{SO}_3^{2-}] > [\text{H}_2\text{SO}_3]$; E) $[\text{SO}_3^{2-}] = [\text{HSO}_3^-] > [\text{H}_2\text{SO}_3]$

Note: Correct option: A - $[\text{H}_2\text{SO}_3] > [\text{HSO}_3^-] = [\text{SO}_3^{2-}]$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'N₂(g)'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 62.2 kJ mol^{-1} . Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Precipitation of CuS. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key